

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
IPDE alpha = 0	31.3	10.3	1.5	0.5	0.2	56.2
IPDE alpha = 0.3	30.1	10	0.4	0.2	0.1	59.2
IPDE alpha = 0.6	27.2	10.4	0.2	0	0	62.2
IPDE alpha = 0.9	23.5	10.2	0.1	0	0	66.2
Mean patients treated						
IPDE alpha = 0	15.4	3.9	2.1	0.9	0.4	
IPDE alpha = 0.3	15.8	3.9	1.8	0.6	0.2	
IPDE alpha = 0.6	16.4	3.7	1.3	0.3	0.1	
IPDE alpha = 0.9	16.9	3.1	1	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
IPDE alpha = 0	34.9	16	31.9	11.6	3.8	1.8
IPDE alpha = 0.3	35.2	19.3	31.9	9.5	2.2	1.9
IPDE alpha = 0.6	36	22.8	30.5	7.5	1.3	1.9
IPDE alpha = 0.9	37.6	26.7	27.2	6	0.5	2
Mean patients treated						
IPDE alpha = 0	11.4	5.1	7.2	3.9	2.2	
IPDE alpha = 0.3	11.9	5.4	7.3	3.7	1.5	
IPDE alpha = 0.6	12.5	6	6.9	3.2	1.1	
IPDE alpha = 0.9	13.2	6.5	6.4	2.8	0.9	
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
IPDE alpha = 0	17.5	40.4	21.1	8.2	2.8	10
IPDE alpha = 0.3	18.4	43.8	19.3	6.8	1.5	10.2
IPDE alpha = 0.6	18.7	46.3	17.9	6.2	0.5	10.4
IPDE alpha = 0.9	18.6	47.4	17.5	5.3	0.4	10.8
Mean patients treated						
IPDE alpha = 0	9.7	7.5	6.1	3.6	2	
IPDE alpha = 0.3	10.1	7.9	5.9	3.4	1.5	
IPDE alpha = 0.6	10.8	8.3	5.6	3	1	
IPDE alpha = 0.9	11.4	8.6	5.3	2.6	0.8	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
IPDE alpha = 0	6.8	7.4	21.4	42.6	19.9	1.9
IPDE alpha = 0.3	7.9	8.5	23.2	41.6	16.8	2
IPDE alpha = 0.6	8.8	9.5	23.9	44	11.9	1.9
IPDE alpha = 0.9	8.8	11.8	24.4	42.7	10.3	2
Mean patients treated						
IPDE alpha = 0	4.5	4.2	6.5	8	6.5	
IPDE alpha = 0.3	4.7	4.4	6.8	8	5.8	
IPDE alpha = 0.6	4.8	4.7	7.1	8.1	4.9	
IPDE alpha = 0.9	4.9	5.1	7.2	8.1	4.3	
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
IPDE alpha = 0	4.9	35.3	26.3	18.1	11.2	4.2
IPDE alpha = 0.3	4.7	34.9	26.3	20.4	9.5	4.2
IPDE alpha = 0.6	4.4	35.6	26.3	21.5	8	4.2
IPDE alpha = 0.9	5	37.9	24.4	21.8	6.7	4.2
Mean patients treated						
IPDE alpha = 0	5	6.4	6.7	5.7	5.7	
IPDE alpha = 0.3	5	6.5	6.9	5.8	5.1	
IPDE alpha = 0.6	5.2	7	6.7	6	4.5	
IPDE alpha = 0.9	5.2	7.2	6.7	6.1	4.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
IPDE alpha = 0	67.6	14.2	3.2	1.2	0.5	13.3
IPDE alpha = 0.3	69.3	12.7	2.8	0.8	0.3	14.1
IPDE alpha = 0.6	69.2	12.9	2.6	0.6	0.1	14.6
IPDE alpha = 0.9	70.3	12.7	1.8	0.2	0	15
Mean patients treated						
IPDE alpha = 0	19.6	3.9	2.5	1.1	0.4	
IPDE alpha = 0.3	20.1	4	2.1	0.9	0.3	
IPDE alpha = 0.6	20.9	4	1.7	0.6	0.1	
IPDE alpha = 0.9	21.9	3.7	1.4	0.3	0	

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IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
IPDE alpha = 0	26.4	36.6	20.5	10.6	3.7	2.2
IPDE alpha = 0.3	26.2	38.5	20.1	9.8	3.1	2.3
IPDE alpha = 0.6	27.3	44.6	17	6.8	1.9	2.4
IPDE alpha = 0.9	27.9	45.9	15.5	7.4	0.8	2.5
Mean patients treated						
IPDE alpha = 0	10.8	7	5.6	3.8	2.5	
IPDE alpha = 0.3	11.1	7.3	5.6	3.6	2	
IPDE alpha = 0.6	11.8	7.8	5.2	3.2	1.5	
IPDE alpha = 0.9	12.4	8.1	5	2.9	1.2	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
IPDE alpha = 0	43.2	9.2	2.8	1.2	0.3	43.3
IPDE alpha = 0.3	43.2	8.4	1.7	0.6	0.1	46
IPDE alpha = 0.6	42.3	8.4	1.2	0.3	0	47.8
IPDE alpha = 0.9	42.6	6.6	0.6	0.2	0	50
Mean patients treated						
IPDE alpha = 0	17.3	3.7	1.9	0.8	0.3	
IPDE alpha = 0.3	18	3.5	1.6	0.5	0.1	
IPDE alpha = 0.6	18.8	3.1	1.1	0.3	0.1	
IPDE alpha = 0.9	19.5	2.7	0.9	0.1	0	
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
IPDE alpha = 0	13.4	10.5	17.5	13.2	26.8	18.6
IPDE alpha = 0.3	15.9	12.7	17.1	13.6	21.2	19.5
IPDE alpha = 0.6	18.6	17.7	16.6	11.8	15.5	19.8
IPDE alpha = 0.9	19.9	19.5	17.6	11.3	11.4	20.3
Mean patients treated						
IPDE alpha = 0	8.9	4.3	5.3	4	5.3	
IPDE alpha = 0.3	9.2	4.6	5.5	3.8	4.5	
IPDE alpha = 0.6	9.9	5.1	5.5	3.7	3.4	
IPDE alpha = 0.9	10.6	5.6	5.2	3.3	2.6	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
IPDE alpha = 0	11.4	31.7	11.8	4.6	2.4	38.1
IPDE alpha = 0.3	12.6	31	11.1	3.7	1.5	40.1
IPDE alpha = 0.6	13	33.4	8.4	2.4	0.8	42
IPDE alpha = 0.9	14.5	31.8	7.4	1.4	0.4	44.5
Mean patients treated						
IPDE alpha = 0	11.7	5.8	4.1	2.2	1.2	
IPDE alpha = 0.3	12.5	5.9	3.8	1.8	0.8	
IPDE alpha = 0.6	13.2	6	3.3	1.5	0.4	
IPDE alpha = 0.9	14.1	6	2.8	1	0.3	
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
IPDE alpha = 0	65.9	22	4.6	0.5	0	7
IPDE alpha = 0.3	66.3	23.3	2.7	0.1	0	7.6
IPDE alpha = 0.6	66.4	24	2.1	0.1	0	7.4
IPDE alpha = 0.9	66.3	24.3	1.4	0	0	8
Mean patients treated						
IPDE alpha = 0	18.4	5.8	3.5	1.1	0.2	
IPDE alpha = 0.3	19.2	5.9	3	0.8	0.1	
IPDE alpha = 0.6	20	5.8	2.5	0.5	0	
IPDE alpha = 0.9	20.9	5.7	1.9	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
IPDE alpha = 0	14.2	12.3	10	36	24.2	3.3
IPDE alpha = 0.3	15	14.4	11.3	36.6	19.5	3.2
IPDE alpha = 0.6	16.2	15.5	12.4	36.6	16.1	3.2
IPDE alpha = 0.9	16.5	18.1	13.4	35.3	13.4	3.3
Mean patients treated						
IPDE alpha = 0	5	4.5	5.3	7.3	7.4	
IPDE alpha = 0.3	5.2	4.9	5.6	7.4	6.5	
IPDE alpha = 0.6	5.3	5.4	5.9	7.4	5.6	
IPDE alpha = 0.9	5.4	5.7	6.1	7.6	4.8	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
IPDE alpha = 0	13.8	32.8	13.4	4.6	2	33.4
IPDE alpha = 0.3	14.8	33.2	12	3.5	1.3	35.2
IPDE alpha = 0.6	14.2	35	10.3	2.6	0.6	37.3
IPDE alpha = 0.9	16.7	33.6	8.4	1.9	0.2	39.2
Mean patients treated						
IPDE alpha = 0	11.4	6.1	4.5	2.5	1.2	
IPDE alpha = 0.3	12.3	6.3	4	2.1	0.8	
IPDE alpha = 0.6	12.8	6.5	3.6	1.6	0.5	
IPDE alpha = 0.9	13.8	6.2	3.2	1.1	0.3	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
IPDE alpha = 0	19.6	14.5	12.8	27.8	18	7.3
IPDE alpha = 0.3	22.2	17.6	13.7	26.9	12.1	7.5
IPDE alpha = 0.6	21.9	19.9	15.5	24.9	10.2	7.6
IPDE alpha = 0.9	22.3	25.6	14.1	23.1	7.3	7.6
Mean patients treated						
IPDE alpha = 0	6.7	5.2	5.8	6.1	5.3	
IPDE alpha = 0.3	7.2	5.6	5.9	5.8	4.5	
IPDE alpha = 0.6	7.6	6.1	6	5.5	3.8	
IPDE alpha = 0.9	7.8	6.6	6.3	5.3	2.9	
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
IPDE alpha = 0	52.4	24.9	10.9	3.6	1.2	7
IPDE alpha = 0.3	51.4	27.9	9.6	3.1	0.6	7.4
IPDE alpha = 0.6	49.2	30.7	9.3	3.1	0.3	7.4
IPDE alpha = 0.9	48.6	34.3	7.5	1.8	0.2	7.6
Mean patients treated						
IPDE alpha = 0	11.6	6.6	5.9	3.3	1.7	
IPDE alpha = 0.3	11.9	7.3	5.6	3	1.2	
IPDE alpha = 0.6	12.4	8	5.3	2.5	0.8	
IPDE alpha = 0.9	13.1	8.2	5	2.1	0.6	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
IPDE alpha = 0	31.4	23.2	16.2	16.6	12.5	0.1
IPDE alpha = 0.3	31	25.4	16.3	16.7	10.5	0.1
IPDE alpha = 0.6	29.3	29.9	15.9	15.8	9	0.1
IPDE alpha = 0.9	27.8	31.3	15.9	17.4	7.5	0.1
Mean patients treated						
IPDE alpha = 0	6.1	5.4	5.4	5.8	7.3	
IPDE alpha = 0.3	5.8	5.7	5.5	5.9	7	
IPDE alpha = 0.6	5.8	5.9	5.6	6	6.7	
IPDE alpha = 0.9	5.6	6.2	5.5	6.2	6.4	
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
IPDE alpha = 0	30.3	49.6	17.9	1.9	0	0.3
IPDE alpha = 0.3	29.2	53.2	16.1	1.2	0	0.3
IPDE alpha = 0.6	29.9	54	14.7	1.1	0	0.3
IPDE alpha = 0.9	29.9	58.1	11.1	0.6	0	0.3
Mean patients treated						
IPDE alpha = 0	10.2	9.7	7.3	2.4	0.4	
IPDE alpha = 0.3	10.6	10.3	6.8	2	0.3	
IPDE alpha = 0.6	10.8	10.7	6.6	1.6	0.2	
IPDE alpha = 0.9	11.4	11.1	5.9	1.3	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
IPDE alpha = 0	9.8	7.5	9.2	14.2	58.5	0.8
IPDE alpha = 0.3	8.5	10	11.1	16.4	53.3	0.7
IPDE alpha = 0.6	9.4	11.1	11	19.2	48.6	0.7
IPDE alpha = 0.9	9.3	14.4	11.3	19.8	44.5	0.7
Mean patients treated						
IPDE alpha = 0	3.8	3.8	4.6	5.5	12.2	
IPDE alpha = 0.3	3.8	4	4.8	5.8	11.6	
IPDE alpha = 0.6	3.8	4.1	5	6	11.1	
IPDE alpha = 0.9	3.8	4.4	5.1	6.3	10.4	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
IPDE alpha = 0	20.6	51.9	17.5	1.6	0	8.4
IPDE alpha = 0.3	19.7	54.7	15.8	1.1	0.1	8.6
IPDE alpha = 0.6	22.2	54.4	13.7	0.6	0	9.1
IPDE alpha = 0.9	23	56.1	11.3	0.4	0	9.2
Mean patients treated						
IPDE alpha = 0	9.2	10.2	7	2.2	0.4	
IPDE alpha = 0.3	9.6	10.6	6.7	1.7	0.2	
IPDE alpha = 0.6	10.3	10.9	6	1.4	0.2	
IPDE alpha = 0.9	11.1	11	5.4	1.1	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
IPDE alpha = 0	12.3	44.8	27.7	9.7	1.2	4.3
IPDE alpha = 0.3	11.3	48.2	27	8.3	0.9	4.3
IPDE alpha = 0.6	13.6	49.4	25.7	6.5	0.5	4.3
IPDE alpha = 0.9	12.1	52.2	24.8	6	0.5	4.4
Mean patients treated						
IPDE alpha = 0	5.6	8.7	8.3	4.6	2.2	
IPDE alpha = 0.3	6	9.1	8.4	4.2	1.7	
IPDE alpha = 0.6	6.2	9.7	8.1	3.8	1.4	
IPDE alpha = 0.9	6.4	10.2	8.2	3.4	1	
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
IPDE alpha = 0	59.9	1.6	0	0	0	38.5
IPDE alpha = 0.3	58.9	1.3	0	0	0	39.8
IPDE alpha = 0.6	58	1.6	0	0	0	40.4
IPDE alpha = 0.9	57.1	1.7	0	0	0	41.2
Mean patients treated						
IPDE alpha = 0	20.9	2.1	0.6	0	0	
IPDE alpha = 0.3	21	1.9	0.4	0	0	
IPDE alpha = 0.6	21.4	1.7	0.3	0	0	
IPDE alpha = 0.9	21.4	1.5	0.2	0	0	

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IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
IPDE alpha = 0	26.3	55.1	12.4	1.9	0	4.3
IPDE alpha = 0.3	24.3	59.5	10.4	1.1	0	4.7
IPDE alpha = 0.6	25.5	58.8	10.6	0.5	0	4.6
IPDE alpha = 0.9	25.3	59.8	9.9	0.4	0	4.6
Mean patients treated						
IPDE alpha = 0	8.7	12.1	6.5	1.7	0.3	
IPDE alpha = 0.3	9.2	12.7	5.9	1.3	0.2	
IPDE alpha = 0.6	9.9	12.9	5.3	1	0.1	
IPDE alpha = 0.9	10.8	12.7	4.9	0.8	0	
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
IPDE alpha = 0	19.6	5.8	0.2	0	0	74.4
IPDE alpha = 0.3	18.1	6.7	0	0	0	75.2
IPDE alpha = 0.6	17	6	0	0	0	77
IPDE alpha = 0.9	15.1	5.8	0	0	0	79.1
Mean patients treated						
IPDE alpha = 0	17.2	2.7	0.6	0	0	
IPDE alpha = 0.3	17.3	2.3	0.4	0	0	
IPDE alpha = 0.6	17.4	2.1	0.3	0	0	
IPDE alpha = 0.9	17.6	1.7	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
IPDE alpha = 0	0.9	1.9	4.8	43.8	48.5	0.1
IPDE alpha = 0.3	1.1	1.9	6	48.1	42.8	0.1
IPDE alpha = 0.6	1.2	1.5	5.6	52.4	39.3	0
IPDE alpha = 0.9	1.2	1.8	5.5	53.6	37.9	0
Mean patients treated						
IPDE alpha = 0	3.3	3.3	3.8	8.2	11.4	
IPDE alpha = 0.3	3.3	3.2	3.8	8.7	11.1	
IPDE alpha = 0.6	3.2	3.2	3.8	9	10.8	
IPDE alpha = 0.9	3.2	3.2	3.7	9.3	10.6	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
IPDE alpha = 0	34	28.3	32.3	4.7	0	0.7
IPDE alpha = 0.3	34.7	31	28.8	4.8	0	0.7
IPDE alpha = 0.6	32.8	35.8	26.6	4.2	0	0.6
IPDE alpha = 0.9	33.5	35.7	27.7	2.4	0	0.7
Mean patients treated						
IPDE alpha = 0	9.3	7.8	9.4	2.8	0.5	
IPDE alpha = 0.3	9.7	8.5	8.9	2.4	0.4	
IPDE alpha = 0.6	10.1	9.3	8.2	2	0.2	
IPDE alpha = 0.9	10.5	9.8	7.7	1.7	0.2	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
IPDE alpha = 0	29	26.5	28.5	5.8	0.1	10.1
IPDE alpha = 0.3	30.7	30.1	24.3	4.3	0	10.6
IPDE alpha = 0.6	31.1	32.4	21.8	3.8	0	10.9
IPDE alpha = 0.9	33.3	34.5	17.6	3.4	0	11.2
Mean patients treated						
IPDE alpha = 0	11	6.5	8	2.8	0.5	
IPDE alpha = 0.3	11.6	7.1	7.5	2.3	0.3	
IPDE alpha = 0.6	12.2	7.7	6.7	1.9	0.2	
IPDE alpha = 0.9	12.9	7.8	6.2	1.5	0.1	
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
IPDE alpha = 0	21.2	21.2	41.9	9.2	1	5.5
IPDE alpha = 0.3	21.2	25.8	39.7	7.1	0.7	5.5
IPDE alpha = 0.6	21.8	28.8	37.3	6.4	0.2	5.5
IPDE alpha = 0.9	23.1	31.8	34.2	5.3	0.1	5.5
Mean patients treated						
IPDE alpha = 0	8.3	6.3	9.4	4	1.4	
IPDE alpha = 0.3	8.5	6.9	9.2	3.6	1.1	
IPDE alpha = 0.6	8.9	7.4	9	3.1	0.8	
IPDE alpha = 0.9	9.4	8	8.7	2.7	0.6	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
IPDE alpha = 0	21.6	37.6	6.8	0.3	0	33.7
IPDE alpha = 0.3	22.5	38.1	4.3	0	0	35.1
IPDE alpha = 0.6	21.8	38.7	3	0	0	36.5
IPDE alpha = 0.9	22.2	36	2.3	0	0	39.5
Mean patients treated						
IPDE alpha = 0	12	8.4	4.3	0.9	0.1	
IPDE alpha = 0.3	12.9	8.3	3.5	0.6	0.1	
IPDE alpha = 0.6	13.6	8.1	2.9	0.5	0	
IPDE alpha = 0.9	14.5	7.5	2.5	0.3	0	
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
IPDE alpha = 0	69.7	23.3	4.1	0.3	0	2.6
IPDE alpha = 0.3	68.5	25.2	3.2	0.1	0	3
IPDE alpha = 0.6	67.9	26.7	2.4	0	0	3
IPDE alpha = 0.9	67.4	27.7	1.9	0.1	0	2.9
Mean patients treated						
IPDE alpha = 0	18.2	7.1	3.6	0.5	0	
IPDE alpha = 0.3	18.4	7.5	3.1	0.4	0	
IPDE alpha = 0.6	19.2	7.3	2.7	0.3	0	
IPDE alpha = 0.9	19.8	7	2.3	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
IPDE alpha = 0	21	13.2	15.7	37	12.3	0.8
IPDE alpha = 0.3	20.3	16.2	14.9	36.4	11.4	0.8
IPDE alpha = 0.6	19	19.6	16.6	35.9	8.1	0.8
IPDE alpha = 0.9	18.3	21	18.3	35	6.6	0.8
Mean patients treated						
IPDE alpha = 0	5.2	4.5	5.7	8.1	6.5	
IPDE alpha = 0.3	5.4	4.8	6.1	8.1	5.6	
IPDE alpha = 0.6	5.4	5.1	6.4	8.1	4.9	
IPDE alpha = 0.9	5.4	5.5	6.7	8.2	4.1	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
IPDE alpha = 0	7.6	35	25.6	25.2	6.5	0.1
IPDE alpha = 0.3	7	34.7	28	24.1	6.1	0.1
IPDE alpha = 0.6	5.6	35.9	27.6	24.7	6.1	0.1
IPDE alpha = 0.9	4.8	36.6	27.5	25.6	5.4	0.1
Mean patients treated						
IPDE alpha = 0	4.1	5.7	6.4	7.7	6.1	
IPDE alpha = 0.3	4.1	5.9	6.5	7.8	5.7	
IPDE alpha = 0.6	4.1	5.9	6.7	8	5.3	
IPDE alpha = 0.9	4.1	6	6.7	8.2	4.9	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
IPDE alpha = 0	9.5	13.2	35	27.3	14.4	0.6
IPDE alpha = 0.3	8.9	14.9	35.8	28.6	11.2	0.6
IPDE alpha = 0.6	9.6	16.5	33.5	30.3	9.5	0.6
IPDE alpha = 0.9	9	17.1	32.7	32.4	8.2	0.6
Mean patients treated						
IPDE alpha = 0	4	4.3	7.6	7	7	
IPDE alpha = 0.3	4	4.3	7.7	7.3	6.6	
IPDE alpha = 0.6	3.9	4.5	7.9	7.7	6	
IPDE alpha = 0.9	3.9	4.7	7.8	8	5.6	
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
IPDE alpha = 0	7.3	5.6	12.7	40	34.3	0.1
IPDE alpha = 0.3	7.2	6.5	13.4	42.7	30.1	0.1
IPDE alpha = 0.6	7	8	15	44.5	25.5	0
IPDE alpha = 0.9	6.7	9	15.6	47.2	21.5	0
Mean patients treated						
IPDE alpha = 0	3.5	3.6	5.2	8	9.7	
IPDE alpha = 0.3	3.5	3.7	5.3	8.4	9.2	
IPDE alpha = 0.6	3.5	3.7	5.5	8.6	8.7	
IPDE alpha = 0.9	3.5	3.7	5.7	9	8	

Phase I/II Operating Characteristics

IPDE alpha comparison - allocation = two_stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
IPDE alpha = 0	13.4	12.2	47.8	18.4	6.2	2
IPDE alpha = 0.3	15	14.2	48	16.3	4.5	2
IPDE alpha = 0.6	15.8	16.6	49	13.2	3.4	2
IPDE alpha = 0.9	15.7	20.8	46.7	12.7	2.1	2
Mean patients treated						
IPDE alpha = 0	5.3	5.6	10	5.3	3.5	
IPDE alpha = 0.3	5.5	6.1	9.8	5.4	2.9	
IPDE alpha = 0.6	5.7	6.7	10.1	5	2.2	
IPDE alpha = 0.9	5.9	7.1	9.9	4.7	1.9	
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
IPDE alpha = 0	12.8	39.8	31.2	11.9	4.2	0.1
IPDE alpha = 0.3	11.8	40.7	33.8	11	2.6	0.1
IPDE alpha = 0.6	11.4	45.1	31.2	10.6	1.6	0.1
IPDE alpha = 0.9	10	47.4	31.4	9.8	1.3	0.1
Mean patients treated						
IPDE alpha = 0	4.7	8.2	8.9	5	3.2	
IPDE alpha = 0.3	4.6	8.7	8.9	5.1	2.7	
IPDE alpha = 0.6	4.6	9.1	9	4.9	2.2	
IPDE alpha = 0.9	4.6	9.8	9.3	4.5	1.8	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
IPDE alpha = 0	20.4	40.2	15.9	6.3	2.3	14.9
IPDE alpha = 0.3	20.6	44.5	12.8	5	1.1	16
IPDE alpha = 0.6	22.1	46.8	11.2	2.8	0.9	16.2
IPDE alpha = 0.9	22.4	49.3	8.9	1.7	0.4	17.3
Mean patients treated						
IPDE alpha = 0	9.6	8.2	5.8	2.8	1.4	
IPDE alpha = 0.3	10.3	8.7	5.3	2.5	0.9	
IPDE alpha = 0.6	11.1	8.9	5	2	0.7	
IPDE alpha = 0.9	11.7	9	4.6	1.5	0.4	
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
IPDE alpha = 0	26.2	49.1	14.1	3.1	0.4	7.1
IPDE alpha = 0.3	24.9	54	11.2	2.2	0.3	7.4
IPDE alpha = 0.6	25.6	56.6	9.3	1.3	0.1	7.1
IPDE alpha = 0.9	26.8	57.3	7.8	0.6	0.1	7.4
Mean patients treated						
IPDE alpha = 0	8.7	10.7	6.1	2.4	0.9	
IPDE alpha = 0.3	9.2	10.9	6	2	0.6	
IPDE alpha = 0.6	9.8	11	5.6	1.6	0.4	
IPDE alpha = 0.9	10.7	11.1	5.1	1.3	0.2	